PRACTICAL PROJECT · MIINA HÄRMA GYMNASIUM · TARTU 2025

Tesla Coil

Construction & Operation

Kristjan Sarapuu & Roomet Tapfer

Supervised by Jörgen Metsik (Chemistry Teacher)

High-Voltage Electronics | 220V AC Input | Spark Gap Design | Self-Built | KR Engineering

€282 Total Cost	~4 cm Spark Length	220V Power Input	2-Storey Design

A high-school practical project in which two students designed, built, and documented a fully functional spark-gap Tesla coil from scratch — achieving 4 cm arc discharges and wireless LED illumination using only a standard 220 V household socket.

OVERVIEW

Project Background & Motivation

This project grew out of a sustained interest in electrophysics developed throughout secondary school. Inspired by YouTube educator Styropyro — known for high-voltage demonstrations that combine rigorous physics with accessible presentation — the two authors decided to build a working Tesla coil as their upper-secondary practical project (UPT).

The decision to collaborate was deliberate: shared costs, complementary skills, and the opportunity to approach problems from multiple angles. A budget ceiling of €200 was agreed at the outset, though the final build ultimately exceeded this by €82 due to component upgrades made during the iterative development process.

PROJECT GOALS

- Construct a functional Tesla coil capable of producing visible arc discharges.
- Design the system around a standard 220 V household socket — no industrial three-phase power.
- Produce a comprehensive bilingual (Estonian/English) safety and operations manual.
- Document the full design, build, and testing process in a formal academic report.

OUTCOMES

- Fully operational spark-gap Tesla coil producing ~4 cm arc discharges.
- Wireless LED illumination demonstrated via electromagnetic induction.
- Custom control panel with safe on/off switching and circuit-breaker protection.
- Safety manual (Version 1, Dec 2024) under the brand name KR Engineering.

THEORY

How a Tesla Coil Works

A Tesla coil — invented by Nikola Tesla in 1891 — is a resonant transformer that converts mains electricity into very high-frequency, very high-voltage alternating current. Tesla originally envisioned arrays of such devices providing wireless power distribution across continents, but the technology proved too inefficient for bulk energy transfer. Today Tesla coils are used almost exclusively for scientific demonstration.

OPERATING PRINCIPLE

Mains AC is stepped up by a high-voltage transformer and used to charge a capacitor bank. Once the capacitor voltage exceeds the breakdown threshold of the spark gap, it discharges in a high-frequency burst through the primary coil (L1). This oscillating current creates a rapidly changing magnetic field that induces a far higher voltage in the secondary coil (L2), which has many more turns. The resulting electric field at the toroid terminal is strong enough to ionise surrounding air, producing visible arc discharges.

An important safety property of Tesla coils is their high impedance: although the output voltage is extreme (tens of thousands of volts), the current that can flow through a human body is very small, and the frequency is high enough that it does not significantly stimulate nerve tissue. Direct contact with the output is therefore survivable — provided the person is not grounded and carries no electronic medical devices.

Key Components

COMPONENT	ROLE IN THE CIRCUIT
High-Voltage Transformer	Steps 220 V mains up to ~4 kV to charge the capacitor bank. Sourced from a salvaged microwave oven.
Capacitor Bank	Stores charge and releases it in high-frequency bursts through the spark gap. Three capacitors in parallel, €
Spark Gap	Air-gap switch: when voltage exceeds ~4 kV, it fires — discharging the capacitors through the primary coil a
Primary Coil (L1)	Low-turn flat spiral coil. Creates the oscillating magnetic field that drives the secondary. ~€50 in copper wire
Secondary Coil (L2)	Tightly wound fine copper wire on a PVC tube. Many more turns than L1 — responsible for the voltage step-
Toroid	Toroidal terminal at the top of L2. Concentrates the electric field and improves arc quality.

COMPONENT	ROLE IN THE CIRCUIT
Mineral Oil Bath	The transformer is submerged in mineral oil inside a sealed container to prevent internal short circuits.
Control Panel	Custom-built box with an emergency stop button, circuit breakers, and output connectors for safe operation.

CONSTRUCTION

Build Process & Challenges

Component procurement began on schedule in mid-2024. Certain parts — particularly the capacitors — arrived only in early August, compressing the build window. The main construction sessions took place at Roomet's family property near Otepää, which offered the necessary tools, workspace, and safe distance from inhabited areas.

Build Timeline

PERIOD	ACTIVITY	KEY OUTCOME
Feb 2024	Project scoping & supervisor confirmed	Budget set at €200; Jörgen Metsik agreed to supervise
May–Jun 2024	Component research & ordering	Google Sheets BOM created; orders placed
Aug 8–9	First prototype assembly	Secondary coil wound; first power-on attempted
Aug 8–9	Troubleshooting — transformer failure	Mains fuse tripped; likely internal short circuit
Aug 8–9	Replacement transformer sourced	Salvaged MOT (microwave oven transformer) — free of charge
Aug 8–9	Second build attempt	Wiring error corrected; coil successfully energised — LED lit wirelessly
Oct–Nov 2024	Intensive rebuild — full redesign	Two-storey frame; copper crimp connectors; mineral oil bath; new windings; cor
Dec 2024	Documentation & safety manual	Bilingual user manual v1 completed under KR Engineering brand
Jan 2025	Report finalised & submitted	25-page written report submitted to Miina Härma Gymnasium

Notable Engineering Decisions

The most significant iterative improvement was the transition from soldered connections to copper crimp connectors throughout the circuit. This dramatically reduced contact resistance and improved arc performance. The transformer was also relocated into a sealed enclosure filled with mineral oil to eliminate the risk of internal arcing.

The two-storey structural frame — fabricated from MDF panels — separates high-voltage components (lower floor: capacitors, spark gap, oil-cooled transformer) from the resonator assembly (upper floor: primary coil, secondary coil, toroid). A grounding conductor runs through both floors and connects to the secondary coil base.

Build Photography

Fig. 1 — First working prototype (August 2024)

Fig. 2 — Revised build nearing completion

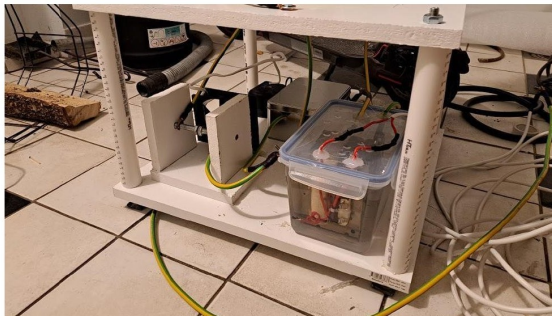

Fig. 3 — Lower floor: capacitors, spark gap & oil-cooled transformer

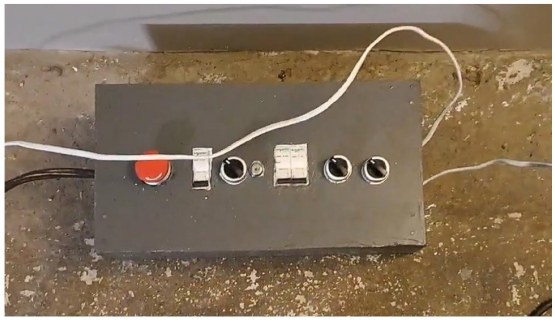

Fig. 4 — Custom control panel with emergency stop

Fig. 5 — Wireless LED illumination via induction

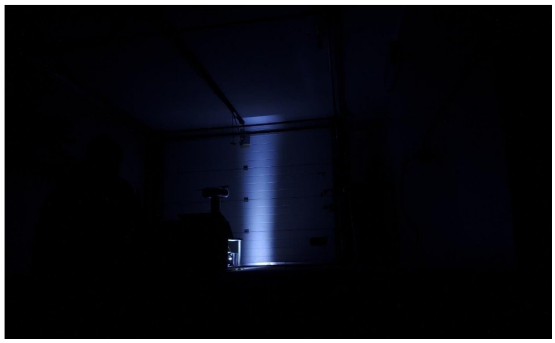

Fig. 6 — Final coil operating in a darkened garage

EVALUATION

Performance Against Assessment Criteria

The project was evaluated against four weighted criteria defined at the planning stage. Each criterion was scored on a 1–5 scale, with descriptions calibrated to the specific technical and safety requirements of high-voltage equipment.

CRITERION	SCORE	BENCHMARK AT THIS LEVEL	ACHIEVED
Power / Arc Length	**4 / 5**	Arc discharges of 4 cm or longer are visible.	~4 cm arcs produced when a grounded metal object is brought

CRITERION	SCORE	BENCHMARK AT THIS LEVEL	ACHIEVED
Safety	**4 / 5**	Device can be operated safely after reading the manual; supervised use required.	Clear labelling; supervised use required. No special training needed,
Practicality	**5 / 5**	Easy to operate; requires only a standard 220 V socket.	Two standard 220 V plugs required; no external ground connec
Visual Finish	**4 / 5**	Painted/finished surfaces; cables neatly routed within enclosures.	Frame painted; cables tidy; lower floor partially open for c

OVERALL SCORE	4 / 5

REFLECTIONS

Individual Evaluations

KRISTJAN SARAPUU

The overall experience was highly positive. We achieved our primary objective while developing knowledge that goes well beyond the standard high-school physics curriculum — particularly around frequency, impedance, and resonance in high-voltage systems.

The main shortcoming was time management: starting the physical build earlier would have allowed more scope for increasing arc power and improving the visual finish. The budget overrun of €82 was a direct consequence of the compressed timeline, which prevented us from sourcing some components second-hand.

This project provides a meaningful foundation for university-level study in electrical engineering or physics, as the concepts involved — resonant circuits, impedance matching, electromagnetic induction — are typically introduced only in the first or second year of an undergraduate programme.

ROOMET TAPFER

Although my initial instinct was to pursue a chemistry-based project, the Tesla coil proved equally absorbing and technically demanding. The experience of iterating through two complete prototypes — diagnosing failures and implementing targeted improvements each time — was genuinely instructive.

The budget overrun could have been avoided with an earlier start, which would have created the opportunity to source components from the second-hand market. Despite this, the final device is a fully operational piece of high-voltage equipment suitable for almost any demonstration context.

The project has generated a desire to continue developing the coil beyond the school submission — specifically to increase output power and improve arc self-ignition. Replacement parts have already been sourced for this next phase.

CONCLUSION

Summary & Key Learnings

The project successfully delivered a functional spark-gap Tesla coil that meets all core performance benchmarks: visible arc discharges, wireless illumination via induction, safe operation from a standard household socket, and a documented safety protocol. Both authors assign the project an overall grade of 4 out of 5.

The primary areas for future improvement are arc power (limited by the current spark gap and capacitor configuration) and structural enclosure of the lower floor. Both improvements have been planned and components are already in hand.

The most valuable academic outcome was the hands-on engagement with resonance, impedance, and high-frequency electrical phenomena — topics that are rarely explored at this depth in secondary education and that provide a genuine head-start for university-level study in electrical engineering or applied physics.

Final Budget Breakdown

ITEM	COST (€)	NOTES
Capacitor bank (3×)	64.00	Ordered online; arrived early August 2024
Primary coil copper wire	50.00	First prototype primary winding
Copper crimp connectors	30.00	Upgrade during October–November rebuild
Wiring & cable	60.00	Full rewire for final build
Spark gap bolts	20.00	Precision stainless M8 bolts
Frame materials, hardware, misc.	58.00	MDF, paint, fasteners, torroid, misc.
Transformer	0.00	Salvaged from a discarded microwave oven
Secondary coil tube & wire	0.00	Materials already on hand
TOTAL	**282.00**	**Initial budget: €200 · Overrun: €82**

Skills Demonstrated

- High-voltage circuit design (4 kV spark-gap topology)
- Electromagnetic coil winding (primary & secondary)
- Fault diagnosis and iterative hardware debugging
- Precision mechanical fabrication (two-storey MDF frame)
- Safety engineering and technical documentation
- Project management: scheduling, budgeting, supplier coordination
- Bilingual technical writing (Estonian / English)
- Graphic design — logo and brand identity (KR Engineering)

REFERENCES

Eldridge, S. (2025). *Tesla coil*. Encyclopedia Britannica. Retrieved 8 January 2025 from https://www.britannica.com/technology/Tesla-coil

Blikai Electronics. (2024). *Tesla Coil: Overview, Principle & Applications*. Retrieved 19 January 2024 from https://www.blikai.com/blog/components-parts/tesla-coil-overview-principle-applications

This document is a portfolio presentation of a practical high-school project. High-voltage equipment — replicate only with appropriate safety precautions and supervision.